Vocabulary resonance in US trademark filings:
A measurement construct for commercial novelty, with validation and applications

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Abstract

The DeDeo prospective/retrospective Kullback–Leibler resonance framework (Barron et al., 2018; Murdock et al., 2017) was developed on cognitive corpora in which the author of each document is the agent whose novelty is being measured: Darwin's reading notebooks, parliamentary speeches of the French Revolution. This paper applies the framework to a qualitatively different corpus type, the goods/services descriptions of US trademark filings, and introduces the resulting construct, denoted ΔKL, as a filing-level measure of commercial novelty that is cheap to compute, available for millions of firms that never patent, and empirically distinct from patent-track invention.

The construct is scored on each filing's topic distribution rather than its raw token counts, following the source papers' own operationalization; this gives every filing a dense representation regardless of length, which matters because goods/services descriptions typically run 10–30 words. The maintenance and firm-level analyses condition on completed registration: completing registration mostly reflects whether the filer followed through to actual commercial use (nine in ten failed applications die by abandonment, not by refusal), so the conditioning separates businesses that launched from filings that remained intentions.

Three results anchor the paper. First, ΔKL marks commercial risk at the mark level, robustly across scoring choices and cohorts. Marks written in language their industry was moving toward die more often at the Section 8 use-proof gates: event-dating every cancellation in the 13.99-million-case-file prosecution record, first-gate failure rises +3.9pp across ΔKL quintiles on 4.12 million registrations 2002–2018. The penalty emerged with the 2008–2010 registration cohorts and has persisted in every cohort since; it is larger for self-filed registrations (+6.9pp) than for counsel-represented ones (+2.5pp), larger where claimed use is older (use-based +5.4pp versus intent-to-use +2.2pp), and it attenuates but persists at the year-ten renewal. Applications register as an inverse-U, with the flux-neutral middle following through to use most reliably. Re-scoring with mixed reference windows shows the gate penalty comes from breaking with the established past, not from resembling the future (−7.3pp against −3.2pp). In either scoring, merely being surprising in the moment is not the signal; the information lives in comparing how surprising a filing is against its past with whether its surprising elements are then taken up by the filings that follow. Second, a methodological caution the construct itself surfaces: under term-level scoring, forward-leaning filings appear to mark better firms (higher margins, more public listings), but these firm-level associations dissolve or reverse under distribution scoring at every resolution tried, and the listing gradient dissolves under document-length controls. Term-level text measures can manufacture firm-performance correlations out of drafting style; distribution scoring is the guard. The firm-level claims are therefore not anchored here, and the within-firm

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patent contrast (-0.048σ, $t = -7.9$) is read only as evidence that the construct is distinct from patent-track invention. Third, the construct supports direct measurement of cross-industry idea diffusion: software- and tech-services-origin themes arrive in transport and advertising/retail with 1–13 year lags, theme adoption follows Bass curves with median $q/p \approx 6.5$, and the cross-class flow is strongly asymmetric (software originates 33 of the 72 origin-to-arrival edges and receives only 10).

The paper closes with a research agenda the construct opens: whether ideas that transfer between industries hold their value, whether suppliers to a new theme outperform its deployers, and how demographic concentration within industries relates to where new themes originate. Code and firm-year panels are public at `https://github.com/ericsilver/novelty`.

1 Introduction

Innovation measurement leans heavily on patents. Patent-based measures are well-validated for technical invention but cover a narrow slice of commercial novelty: most firms never patent, service-sector and business-model innovation rarely patents, and patenting propensity varies so much across industries that cross-industry comparisons are fraught (Dinlersoz et al., 2018; Graham et al., 2013). Trademark filings offer a complementary window. Nearly every commercially serious product or service introduction in the United States is accompanied by a trademark filing, the filing's goods/services description states in plain commercial language what the firm intends to sell, and the corpus extends across every sector of the economy at the rate of hundreds of thousands of granted filings per year. The text of those descriptions, however, has been largely unused; the literature has worked with filing counts and filing metadata rather than filing language.

Murdock et al. (2017) introduced a measurement framework for directional novelty in time-ordered text corpora: for each document, compare its token distribution to the aggregated distribution of past documents (prospective surprise) and future documents (retrospective surprise). The signed difference of those two quantities they call *resonance*: it is positive when a document is unusual against its recent past and close to its near future, negative for the inverse. The framework was developed on Charles Darwin's reading notebooks and extended in Barron et al. (2018) to French Revolution parliamentary debates. Both corpora share a salient property: the text is authored by the agent whose novelty is the object of measurement.

This paper applies the same apparatus to trademark goods/services text, which is commercially incentivised, typically drafted by counsel, and exists to define legal protection scope rather than to express cognition. Whether the resonance framework produces interpretable structure on text of this kind is the threshold question, and the answer this paper documents is yes. The construct, denoted ΔKL, marks commercial risk at the level of the individual mark, robustly across scoring choices. Filings whose vocabulary leans toward where their industry's language is going fare worse as legal objects: their applications complete registration slightly less often (a step that mostly measures follow-through to commercial use, since roughly nine in ten failed applications die by abandonment rather than by refusal), and the registered marks fail the first Section 8 maintenance gate more often. The construct also surfaces a methodological hazard that this paper documents rather than falls into: under term-level scoring, the same forward-leaning filings appear to belong to more successful firms, but those firm-level associations dissolve or reverse under distribution-based scoring at every resolution tried, and the listing gradient dissolves under document-length controls. The signal is uncorrelated-to-negatively correlated with patenting within firm; ΔKL captures a dimension of commercial text that the patent record does not, whatever its ultimate relation to firm performance.

Because the construct is computed from public text with public code, it scales to questions that firm-survey or patent-linked measures cannot easily reach. The paper demonstrates one such application in depth, measuring the diffusion of business vocabulary across industries directly at the phrase and theme levels, and sketches a research agenda: the portability of transferred ideas, the relative fortunes of suppliers versus deployers of a new theme, and the relationship between demographic concentration within an industry and where new themes are born.

Section 2 defines the construct, the corpus, and known fragilities. Section 3 reports the commercial-outcome validation. Section 4 reports the within-firm patent contrast and the expert-list discriminant. Section 5 reports the cross-industry diffusion application. Section 6 reports a base-rate discipline result on entrant-vs-incumbent classification. Section 7 states the research agenda. Section 8 bounds the claims. Section 9 is reproducibility.

2 The construct

2.1 The DeDeo apparatus on trademark text

For a time-ordered sequence of documents, with each document at year t producing a token distribution P_t, the framework defines two quantities against a window W of past and future documents (Barron et al., 2018; Murdock et al., 2017):

- *Prospective surprise*: $\mathrm{KL}(P_t \,\|\, Q_{\text{past}})$, where Q_{past} is the aggregate token distribution of documents in years $t - W$ to $t - 1$.
- *Retrospective surprise*: $\mathrm{KL}(P_t \,\|\, Q_{\text{future}})$, where Q_{future} is the aggregate distribution of documents in years $t + 1$ to $t + W$.

The signed resonance is

$$\Delta\mathrm{KL}_t \;=\; \mathrm{KL}(P_t \,\|\, Q_{\text{past}}) - \mathrm{KL}(P_t \,\|\, Q_{\text{future}}).$$

Positive ΔKL marks a document whose vocabulary is unusual relative to its recent past and close to its near future: the field moved toward this filing. Negative ΔKL marks the inverse, language that looked normal at filing and looked played-out in hindsight: the field moved away.

2.2 Corpus and conditioning

The USPTO Trademark Annual Applications backfile (product TRTYRAP) provides a complete record of US trademark filings from 1884 onward. The deepest diagnostic work in this paper uses the complete software/electronics class (NICE 009; 1.81M filings), and the diffusion application uses four classes selected to exercise canonical cross-industry flow: software (009), scientific and technical services (042), transport (039), and advertising and retail (035), totalling 2.32 million granted filings 1990–2024. The firm-level panels link trademark owners to SEC EDGAR financial filers and to PatentsView patent assignees.

Two conditioning rules structure the analyses. The diffusion and theme analyses restrict to *granted* filings (registration date populated), screening out speculative or defensive applications that did not survive examination; the lifecycle analyses necessarily score all applications, since registration completion is itself one of their outcomes. And the lifecycle outcomes in Section 3 keep the examination step and the maintenance step separate: registering and maintaining are different commercial acts with different drivers, and pooling them conflates examiner behavior with the use requirement.

Outcomes are measured two ways. Mark status is read from the USPTO status-code dictionary directly (700 registered; 701–705 Section 8 accepted; 710 cancelled for Section 8 failure; 800 renewed; 900 expired), ground-truthed against marks with publicly known fates; because a status snapshot cannot attribute a cancellation to a specific maintenance gate, snapshot-based maintenance results use registration cohorts for which exactly one gate has elapsed. The primary gate results, however, are *event-dated*: a full re-parse of the TRTYRAP backfile (13.99 million case files) yields each application's prosecution history (office actions, responses, abandonments, and dated Section 8 cancellations), the attorney of record, and the filing basis, so a cancellation is assigned to the first maintenance gate by its date relative to registration, every elapsed cohort is usable, and representation and basis are observed rather than proxied.

2.3 Computation

The construct is scored on topic distributions, following the source papers' own operationalization: the Darwin and French Revolution analyses computed surprise on each document's topic mixture rather than its raw token counts (Barron et al., 2018; Murdock et al., 2017). A latent Dirichlet allocation model with $T = 50$ topics is fit on a stratified sample of 448,437 filings drawn across all 45 classes (vocabulary of 62,168 unigrams and bigrams), every filing 1990–2024 is transformed to a dense 50-topic distribution, and each filing's prospective and retrospective KL are computed against class-year topic references at $W = 5$, scoring 1995–2019. Topic scoring matters here for a reason the source corpora did not face: goods/services descriptions typically run 10–30 words, and on documents that short, token-level KL inflates mechanically (mean $|\Delta\mathrm{KL}|$ varies 2.5-fold across document-size bands), while every filing's topic distribution is dense regardless of length and the artifact largely disappears. A $T = 200$ refit confirms the headline results are not sensitive to topic resolution.

A term-level variant is retained alongside: token distributions (unigram + bigram, within-class document frequency ≥ 50) against the same class-year windows, with Dirichlet smoothing ($\alpha = 1$ per cell), computed at $W \in \{3, 5, 7\}$. The term variant is used where vocabulary identity is the object (the phrase-transit analysis, the reference-window decomposition, the months-scale freshness measure), and as a comparison scoring throughout; because its two windows need not match, it also composes into asymmetric constructs (prospective surprise against one window, retrospective against another) that Section 3.2 uses to ask *which* novelty the outcomes respond to.

The signal depends on having enough past and future filings to populate the references. Figure 1 shows the reference-window density by year for two diagnostic classes; the 5-year past window is incomplete before 1990 and the 5-year future window is incomplete after 2020, and both zones are excluded from the construct's interpretable range. Within the supported window the classes analyzed here have between ~3,500 and ~80,000 filings per year.

The burn-in cut is set empirically rather than by the window arithmetic alone. Thin past references inflate prospective surprise, so contamination appears as a positive bias in the per-year mean of signed $\Delta\mathrm{KL}$ relative to the class's long-run level (profiled under term scoring, where the thin-reference artifact is sharpest). Profiling that bias across all 44 classes with sufficient volume: the median absolute bias is ~2.0 nats in 1990, 0.25 in 1991, 0.10 in 1992, and 0.04 from 1993 onward, below any effect size this paper interprets. The deviations that reappear in 1995–1997 (~0.15–0.17) are not burn-in, because mechanical contamination cannot rebound; they are the dot-com era moving the corpus, which a longer burn-in would discard as data. Per-class burn-in lengths were also estimated under a stability-band criterion; the apparent cross-class differences track each class's noise floor rather than its reference density (correlation of burn-in length with early filing volume +0.26, with a slow-moving-class proxy +0.14), so there is no empirical case for industry-specific burn-ins. All analyses therefore use a single standard cut at 1995, the first scored year whose full 5-year past window lies inside the dense corpus, which the bias profile shows is conservative by about two years.

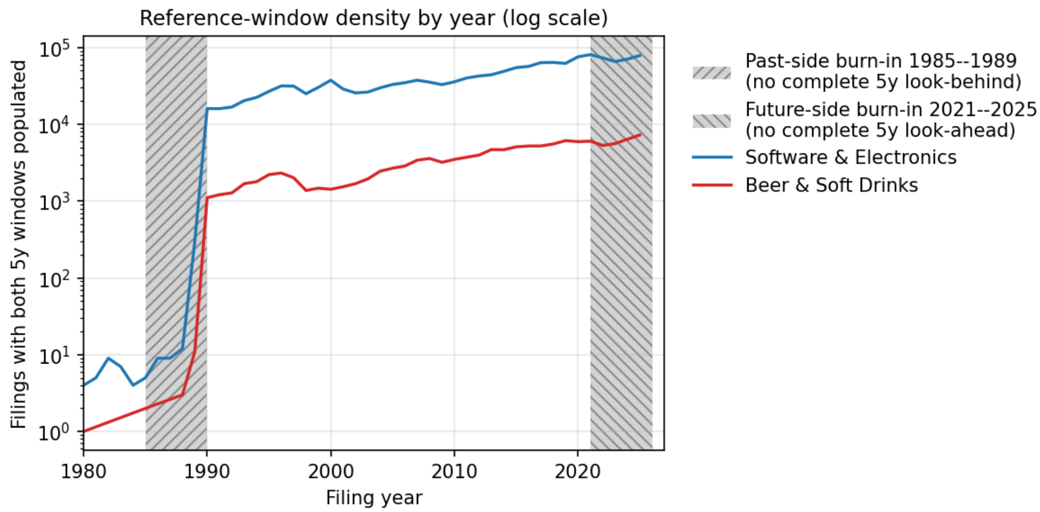

Figure 1: Reference-window density by year on log scale. Hatched regions are the past-side and future-side burn-in zones where the 5-year window is incomplete.

2.4 The construct at the case level

Figure 2 grounds the construct in recognizable examples. The two scatter axes are prospective surprise and retrospective surprise, with ΔKL as the diagonal residual. Filings below the diagonal are *innovator*: the language looked unusual at filing and became standard afterwards. Filings above the diagonal are *tired*: the past resembles the filing more than the future does.

2.5 Known fragilities

The construct has documented fragilities on this kind of text, and they motivate the choice of topic scoring as the method. Prospective and retrospective KL are highly correlated however scored (term scoring: $r = +0.971$ on the complete Class 009 corpus; topic scoring: 0.950), so signed ΔKL is a small residual of two large, stable quantities; what has not been seen before will mostly not be seen again, and the signal lives in the difference. Under term scoring the residual is noisy at the per-filing level, and its magnitude inflates mechanically for short filings (mean $|\Delta\text{KL}|$ of 0.293 for documents under 5 terms, decaying to 0.147 at 40 or more, a 2.5-fold gradient); under topic scoring the gradient falls to ~1.4-fold, which is why topic scoring carries the outcome analyses. The two scorings agree only moderately per filing (Pearson 0.46, Spearman 0.36 on 1.9M jointly scored filing-class observations), which is what gives their agreement on the mark-level outcome patterns evidential force, and their disagreement on the firm-level patterns diagnostic force (Section 3.3). Pre-1995 readings are contaminated by thin past references. ΔKL is robust to the duration of the timing window considered: across $W \in \{3, 5, 7\}$ years, only 6.9–10.4% of filings flip sign, well under conventional fragility thresholds. The per-filing noise attenuates at every aggregation used in this paper: firm-year means, token-level pools, and theme-level prevalence trajectories.

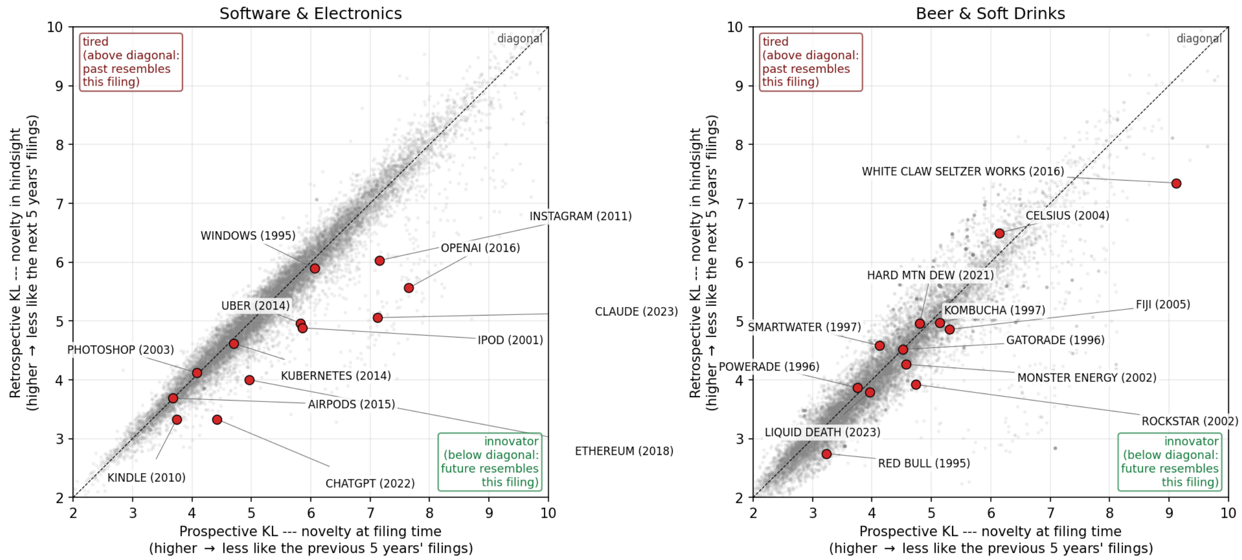

Figure 2: Prospective KL versus retrospective KL with ΔKL as the above-diagonal/below-diagonal residual. Labeled examples are recognizable post-burn-in brand introductions, positioned by their goods/services text. *Left:* software/electronics. OpenAI (+2.09), Claude (+2.06), Instagram, ChatGPT, iPod, Ethereum, and Uber sit on the innovator side; Photoshop (2003) and AirPods sit essentially on the diagonal. *Right:* a beer/soft-drinks diagnostic class. White Claw (+1.78), Rockstar, and Red Bull lean innovator; Smartwater (1997) and Celsius (2004) lean tired.

3 Validation against commercial outcomes

If ΔKL measures something commercially real, it should predict outcomes that cost firms money. At the *mark* level it does, robustly: the trademark lifecycle selects against forward-leaning vocabulary, whose applications complete registration slightly less often and whose registered marks fail the first maintenance gate more often, under topic scoring at two resolutions and under term scoring alike. At the *firm* level the corpus teaches a different lesson. Term-level scoring suggests that firms filing forward-leaning marks do better (listing, margins); distribution-based scoring erases or reverses those associations at every resolution tried. Section 3.3 documents the divergence and reads it as a methodological hazard for text-based novelty measures generally, rather than asserting either sign as a property of firms.

3.1 Mark-level selection: registration and the first maintenance gate

Outcome definitions. A filing reached registration if its registration date is populated. A registered mark faces the Section 8 use-declaration gate in years five to six after registration. The primary gate measure is event-dated: a mark failed the first gate if its prosecution history records a Section 8 cancellation at registration-age four to eight and a half years, which makes every elapsed registration cohort (2002–2018) usable and separates the first gate from the year-ten renewal. The snapshot cross-check reads terminal status instead (*Cancelled–Section 8*, status 710, against *Section 8 accepted*, 701–705, or renewed, 800) and is restricted to the 2016–2018 cohort, for which exactly one gate has elapsed and the two are therefore not confounded.

Registration completion is better read as a launch indicator than as an examiner verdict: among applications that fail to register, roughly nine in ten die by abandonment (44% never file the

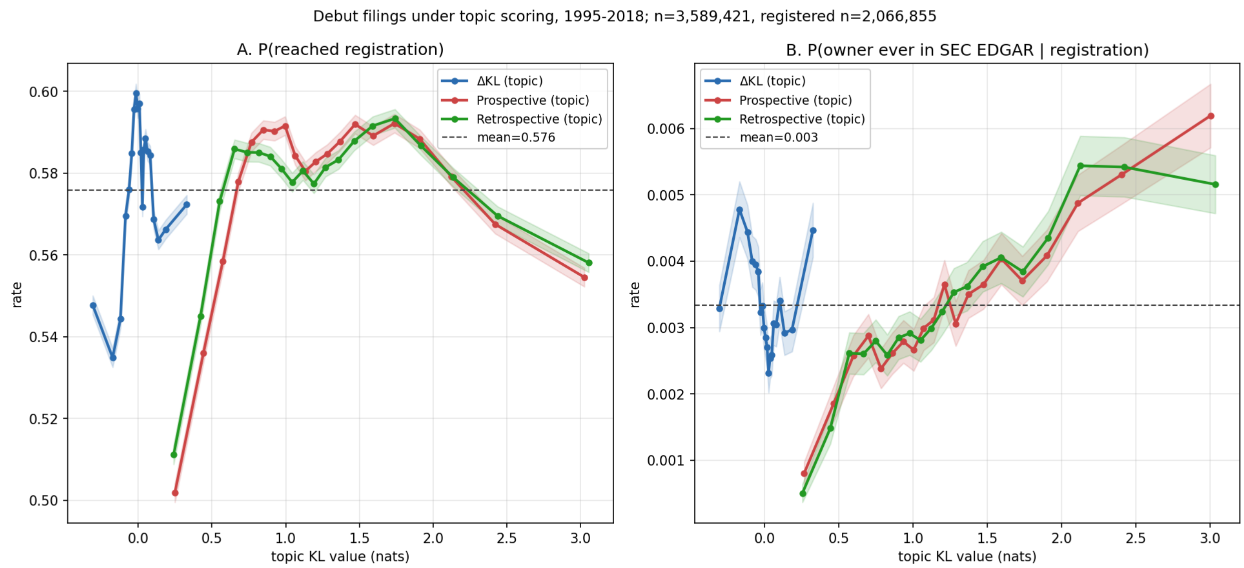

Figure 3: Outcomes for debut filings (owner's first-ever filing) across all 45 NICE classes under topic scoring, filing years 1995–2018, in quantile bins with 95% intervals. $n = 3{,}589{,}421$; registered subset $n = 2{,}066{,}855$. *A:* P(reached registration). *B:* P(owner ever in SEC EDGAR | registration); the absence of a ΔKL gradient in panel B is discussed in Section 3.3.

statement of use that registration requires; 45% stop responding during prosecution) and only a few percent lose adversarial or appeal proceedings. Completion (Figure 3A; 3.59 million debut filings, 57.6% base rate) is an inverse-U in ΔKL: the middle quintiles complete best (∼58.6–58.9%) and both tails complete less (54.5% at the bottom quintile, 56.8% at the top), with the same hump in the topic-KL levels. On the launch-indicator reading, the inverse-U says that applicants filing category-typical language most reliably follow through to commercial use, while both kinds of vocabulary bet fail to launch more often. The shape is scoring-robust (term scoring gives depth +4.8pp on all filings pooled; topic scoring +3.7pp, and +3.0pp at $T = 200$) and is the rule rather than a pooled artifact: estimated class by class under term scoring, the middle-versus-tails depth is positive in 37 of 44 classes with sufficient volume and statistically distinguishable from zero in 33 (per-class figure and table in the repository).

The registration inverse-U is also why analyses that skip the registration conditioning see an inverse-U in downstream outcomes. Any unconditional success measure, mark survival included, is the product of the registration rate and the conditional rate, and the registration factor stamps its middle-favoring shape onto the composite: the flux-neutral middle registers best, so it appears to "survive" best unconditionally even where the conditional gate pattern says otherwise. Appendix A reproduces the outcome analysis without the registration conditioning to make the composite shape visible.

The first Section 8 gate is where forward-leaning marks pay. The event-dated estimate pools every elapsed cohort: among 4.12 million registrations from 2002–2018 across all classes, the probability of a dated Section 8 cancellation at the first gate rises from 43.7% at the bottom topic-ΔKL quintile to 47.7% at the top, a +3.9pp failure lift (SE 0.08pp), positive in 31 of 44 classes with sufficient volume (Figure 5). The snapshot estimate on the single one-gate-elapsed cohort agrees (−5.5pp survival on 2016–2018 registrations) and is scoring-robust: −5.7pp under term scoring on the identical sample, −5.5pp at $T = 200$ (Figure 4). Under term scoring the gate survival *rises* in the raw KL level (+6.5pp prospective, +7.6pp retrospective): distinctive text survives the use requirement better; forward-leaning text survives it worse. A natural mechanism runs through the use declaration itself: Section 8 requires continued use in commerce, and vocabulary that bets on where the category is going belongs disproportionately to products that had not yet materialized at filing and sometimes never do.

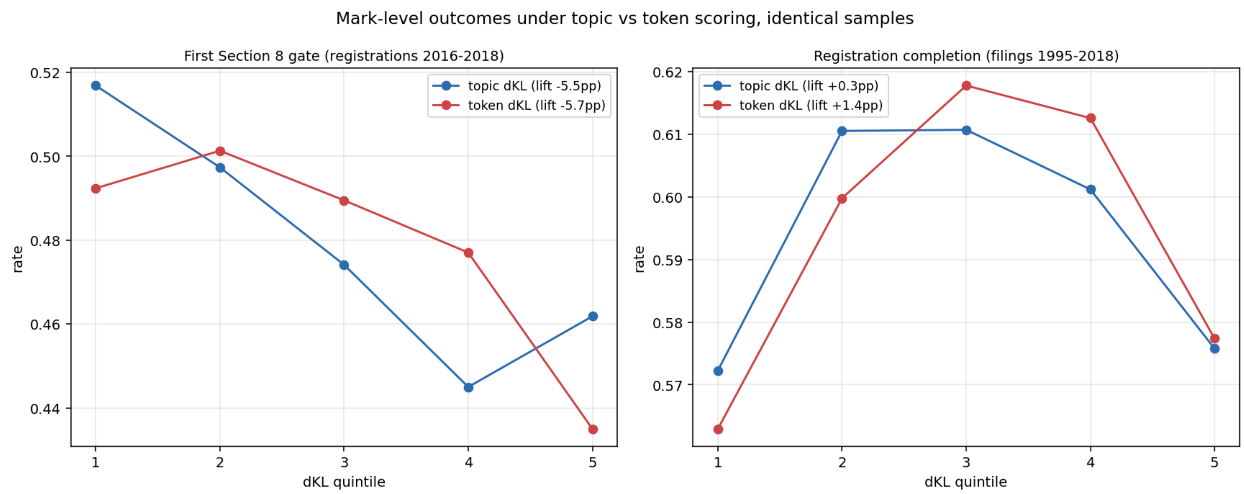

Figure 4: The two mark-level outcomes under topic and term scoring on identical samples, pooled across classes. *Left:* first Section 8 gate survival (snapshot, registrations 2016–2018). *Right:* registration completion (filings 1995–2018). Both patterns are scoring-robust.

The event history adds three facts the snapshot could not see. First, *the penalty is modern*: by registration cohort, the failure lift is indistinguishable from zero for 2002–2007 cohorts, switches on over 2008–2010, and runs +4 to +9pp for every cohort since, largest for the 2018 cohort whose gate closed in 2023–24 (Figure 6). The forward-leaning gate risk is a feature of the post-2008 filing economy, not of trademarks per se, and it predates the pandemic by a decade. Second, *representation buffers the bet*: the lift is +6.9pp among self-filed registrations against +2.5pp among attorney-represented ones (with self-filers also failing the gate far more often overall, 56.8% versus 43.0%). Third, *recent proof of use buffers it too*: intent-to-use registrations, whose statement of use was examined at registration, carry a +2.2pp lift against +5.4pp for use-based registrations, whose claimed use may be older than it looks. Per class, the largest penalties sit in tobacco (+26pp, the vaping era), scientific/tech services (+14pp), and lighting (+12pp, the LED transition), with small negative lifts in staple goods (jewelry, foods, alcohol).

The maintenance risk does not end at the first gate, but it attenuates and changes shape. In the software diagnostic class, where the event-dated first-gate failure lift on the 2010–2013 cohorts reaches +20.8pp, the conditional lift at the second gate (the year-ten combined renewal, among marks that passed the first) falls to +7.0pp; pooled across classes the second-gate pattern is a shallow U (40.4% at the bottom ΔKL quintile, 37.2% in the middle, 42.2% at the top), so by year ten the backward-leaning tail has caught up in mortality. Most of the ambition bet resolves at first proof of use; what remains is a smaller persistent risk on the forward side and, eventually, fashion decay on the backward side. Separating the gates requires the event dates: in the status snapshot, a year-ten cancellation is indistinguishable from a year-six one.

The prosecution history also functions as firm-health telemetry, and it sharpens the mechanism. Among represented filings in the software class, mean response time to office actions predicts the gate: fastest-quintile responders pass at 57.4% against 49.4% for the slowest. The ΔKL penalty survives within both fast and slow halves (-7.2 and -8.0pp), so it is not an artifact of dying firms detected late; the risk attaches to the product bet itself. The telemetry also documents the representation economics directly: counseled deaths occur at a median mark age of 3.2 years against 1.5 for self-filers, more often after registration (38% versus 28% of deaths at the gate rather than in prosecution), and more often after responding to at least one office action first — firms with the resources to retain counsel persist in the record well past the point where a self-filer's death would already have been recorded.

Summarizing the cross-industry picture: the registration inverse-U is near-universal (37 of 44 classes), and the event-dated gate penalty is the majority pattern (31 of 44), with the exceptions concentrated in staple and traditional-goods classes (jewelry, staple foods, construction, beer, alcohol) where forward-leaning vocabulary carries no maintenance risk and modestly helps. The penalty concentrates where forward bets are speculative: emergent-category goods and technical services.

3.2 Which novelty? Decomposing the gate penalty by reference window

"Forward-leaning" is relative to a reference, and the choice of reference window selects what kind of novelty is being measured. Scoring the full corpus at $W \in \{3, 5, 7\}$ and mixing windows across the two sides gives five variants: three symmetric, plus a foresight configuration (prospective against the recent 3-year past, retrospective against the distant 7-year future) and a past-rupture configuration (prospective against the established 7-year past, retrospective against the near 3-year future).

The gate penalty is monotone in the mix (Figure 7). Pooled across classes on the 2016–2018 gate cohort, the top-versus-bottom quintile lift runs -3.2pp for the foresight configuration, -4.7pp at symmetric $W = 3$, -5.7pp at the $W = 5$ baseline, -6.1pp at $W = 7$, and -7.3pp for past-rupture, which is also the most pervasive variant across industries (negative in 35 of 41 classes, against 31 for the baseline). In software the decomposition is complete: the foresight configuration carries no gate penalty at all (-0.3pp) while past-rupture carries -12.7pp. The Section 8 risk attaches to breaking with the established past, not to resembling the future; lengthening the past window concentrates the penalty and lengthening the future window dilutes it. This is consistent with vocabulary leaving the corpus gradually rather than abruptly: the long-past reference holds the fading establishment, and distance from that establishment is what the use requirement punishes. The registration inverse-U, by contrast, is indifferent to the window mix (middle-versus-tails depth $+4.6$ to $+4.9$pp across all five variants), so the examiner and the marketplace are responding to different aspects of the same text.

The decomposition extends below the year. Scoring class 009 at monthly resolution against two adjacent references, the twelve months ending one month before filing and the twelve months before those, the two surprise levels are nearly identical per filing ($r = 0.992$, the stability property again), but their difference, a months-scale freshness measure that is high when a filing resembles the very recent months far more than the year before them, reproduces the risk pattern: marks riding a wave that emerged within the prior year fail the gate at -9.4pp across freshness quintiles, a stronger separation than the symmetric ΔKL baseline delivers in the same class. Two things follow. Month-scale position in a wave matters, with the penalty concentrated on the months-late rather than the months-early. And because freshness is built entirely from pre-filing text, it is computable on the filing date with no look-ahead, which makes it the construct family's real-time member. This is a single-class result and its corpus-wide estimation is owed.

3.3 Firm-level associations are scoring-dependent

Under term-level scoring, forward-leaning filings appear to belong to better firms. Among registered debut filings, the probability that the owner ever appears in the SEC EDGAR listed universe rises in term-ΔKL from 0.29% at the bottom quintile to 0.42% at the top; on an SEC-matched panel of firm-years, trailing 3-year mean term-ΔKL predicts gross margin at +2.3pp per σ ($t = 2.7$ on the locally rebuilt panel through 2019; +3.2pp, $t = 5.3$, on the original panel through 2024), with prospective surprise entering positively and retrospective negatively, the resonance configuration; and firm-year term-ΔKL predicts excess stock returns of +1.5 to +5.0pp per σ at one-to-four-year horizons, decaying by year five, with prospective surprise alone predicting approximately zero at every horizon. A larger debut-only return figure (+28pp at 4 years) does not survive a sample-size diagnostic and is not claimed.

The *signed* versions of these associations do not survive distribution-based scoring, and the paper therefore does not anchor on them. Under topic scoring on identical samples, the debut listing gradient in signed ΔKL is flat to declining (Figure 3B), and the margin coefficient reverses: -1.3pp per σ ($t = -1.8$) at $T = 50$ and -2.2pp ($t = -3.1$) at $T = 200$, with the prospective/retrospective decomposition flipping sign. Quadrupling the topic count moves the estimates further from the term-level result, not toward it; this resolution test holds the topic vocabulary fixed, so it speaks to granularity but not to vocabulary coverage, which is tested separately below. Meanwhile the term-level listing gradient dissolves under document-length controls: within document-size bands it is U-shaped rather than rising, with the apparent monotone rise contributed by composition across lengths (listed firms' counsel write longer descriptions; the only large within-band effect is in 40-plus-term filings). The mark-level results of Section 3.1 are unaffected: they hold under both scorings at both resolutions.

The strongest objection to the artifact reading is vocabulary blindness: the topic model's vocabulary is built from a corpus sample, so rare and emergent terms are invisible to it, and if the firm-level signal lives in exactly those terms, topic scoring would erase a real signal rather than an artifact. The objection is testable, and testing it identifies what the term-level signal actually was. The marquee emergent unigrams (blockchain, cryptocurrency, machine learning) are in the topic vocabulary, and at $T = 200$ they anchor their own topics; the invisible tier is the rare tail (63.8% of class-009 term *types*, but only $\sim$9% of token *mass*, with nft, cbd, fintech, and smartwatch among them). Computing each filing's invisible-vocabulary share gives the discriminating split. The share itself is a strong firm marker: P(owner ever in SEC EDGAR) rises from 2.9% to 5.5% across its quintiles, and gate survival rises +5.7pp. But *within* invisible-share strata, the signed ΔKL gradient on the firm outcome is flat to negative, while the gate penalty holds in both strata (-6.9 and -8.3pp). The term-level firm signal, in other words, is *unsigned specificity*: serious firms describe specific products with specific, therefore rare, words, and rare-term mass inflates term-ΔKL mechanically. What dissolves under topic scoring is the specificity channel as a *signed* signal; the directional resonance claim about firms is what remains, and it is empty.

The unsigned association itself is robust under every scoring, and it is the constructive residue of this section. Under topic scoring, the debut listing probability rises monotonically in the KL *level* on both axes, from 0.20% at the bottom prospective quintile to 0.51% at the top, replicated at $T = 200$ (0.19% to 0.50%) and on the retrospective axis, a 2.5-fold gradient that cannot be a length artifact (the topic levels are nearly length-uncorrelated, and listed firms file long, so any length bias runs against it). Atypical text belongs to listed firms; text that leans *forward* specifically does not belong to them any more than text that leans backward. Lexical distinctiveness, not resonance, is the firm-quality signal in this corpus, and measuring it directly is likely a better instrument than ΔKL for firm-level work.

This is a finding about measurement with reach beyond this corpus: term-resolved text-novelty measures can manufacture firm-performance correlations from drafting style and lexical specificity, and a distribution-based rescoring is a cheap, decisive diagnostic. What does *not* explain the firm-level divergence is the registration filter. Among applications that fail to register, only 4.3% are removed adversarially (appeal, opposition, or inter partes proceedings); 42.6% never file the statement of use and 45.3% stop responding during prosecution, applicant decisions rather than examiner ones. Dropping the registration conditioning entirely leaves the term-level listing gradient essentially unchanged (0.26% to 0.38% across quintiles against 0.29% to 0.42% conditional), so neither the term-level association nor its dissolution is an examiner artifact; and an examiner channel that touches under two percent of applications cannot generate the several-point quintile gaps of the registration inverse-U either.

If a genuine signed firm-level effect exists beneath the artifacts, the panels here cannot establish it, and two causal readings would still compete: a realized-business channel (the trademarked line itself pays) and a propensity channel (firms disposed to bet early both file forward-leaning marks and perform). Four tests would settle the remaining questions and are owed: rescoring topics on the full per-class term vocabulary (removing the sample-truncation objection entirely); a future-blind vintage refit of the topic model (removing the time-pooling objection); decomposing each filing's term-ΔKL into in-vocabulary and out-of-vocabulary contributions and re-estimating the firm panels on each; and an emergence-event design that flags filings carrying a to-be-emergent term in its first corpus years and compares firm outcomes against class-, year-, and length-matched controls with attorney-of-record fixed effects. The never-registered-applications test of the propensity channel completes the list.

4 Distinctness from patents

If ΔKL and patent counts were noisy measures of the same underlying construct, they should correlate within firms over time. They do not.

The within-firm analysis uses a published panel of 174,569 firm-years of trademark owners name-matched to PatentsView patent assignees (PatentsView, 2024); the fixed-effects estimation uses the 94,106 firm-years across 14,470 firms with within-firm variation. The panel and matching code are in the public repository. The matching normalizes both sides (strip corporate-form suffixes and punctuation, lowercase) and joins on exact normalized-name equality. One provenance note: the published panel was constructed under an earlier exponential-decay kernel variant of the construct; its firm-year mean ΔKL correlates at $r = 0.87$ with the current flat-window scoring on 68,013 matched owner-years, and the re-estimation on current scorings follows the table. I fit a two-way (firm + year) fixed-effects regression of standardized firm-year mean ΔKL on $\log(1 + \text{patents})$ with firm-clustered standard errors.

The coefficient is negative and precisely estimated, strengthening under filing-count control. The strongest negative coefficient sits on patents_{t-1}: in years after a firm patents heavily, its trademark vocabulary regresses toward class-typical. A reading consistent with the data is temporal substitution between patent prosecution and market-facing brand work; either way, the two measures are not interchangeable.

Re-scoring the same matched sample under the method sharpens the distinctness claim rather than weakening it. On 89,987 firm-years rebuilt symmetrically from current per-class scores (1995–2019), the contemporaneous two-way-FE coefficient is -0.038σ ($t = -6.2$) under term scoring, -0.014 ($t = -2.3$) under topic scoring, and -0.010 ($t = -1.5$) at $T = 200$: under distribution scoring, ΔKL is even closer to exactly orthogonal to patenting, which is the property the construct

Table 1: Within-firm ΔKL versus patenting. Coefficients in standard deviations of ΔKL per unit $\log(1 + \text{patents})$.

Specification	coef (σ)	SE	t
Between-firm correlation (firm means)	−0.020	—	—
Within-firm correlation (firm+year demeaned)	−0.015	—	—
Two-way FE, contemporaneous	−0.048	0.006	−7.9
robustness: patents = max over matched assignees	−0.048	0.006	−8.0
controlling log(n_filings)	−0.051	0.006	−8.4
Lead/lag: patents$_{t-1}$	−0.054	0.009	−5.8
patents$_t$	−0.019	0.010	−1.9
patents$_{t+1}$	+0.010	0.010	+1.0

needs.

An earlier version of this analysis reported that firms on BCG/MIT "most innovative" lists sit $+0.34\sigma$ above the panel mean in ΔKL ($n = 43$ matched, $p < 0.001$). That result does not replicate on the current pipeline and is withdrawn: rebuilding the firm-level panel symmetrically from current per-class scores gives $+0.09\sigma$ ($p = 0.22$) under term scoring, $+0.14\sigma$ ($p = 0.07$) under topic scoring, and $+0.08\sigma$ ($p = 0.20$) at $T = 200$. The original figure was computed on a panel built under the earlier decay-kernel variant, and the matched samples are small ($n = 41$ here); whatever alignment exists between ΔKL and expert innovation judgment is at most suggestive. Patent counts remain uncorrelated with ΔKL on the same panel (Pearson +0.010).

5 Application: cross-industry diffusion, measured directly

Because ΔKL is computed from open text, the same corpus supports measuring where business vocabulary is born and how it travels. This section demonstrates the construct's reach beyond the lifecycle validation.

5.1 Phrase-level transit

For 14 curated business-model phrases, I record the first year each of the four classes crosses a 1% prevalence floor among granted filings.

Table 2: First year a theme reaches 1% prevalence among granted filings, by class. "–" = never clears the floor.

Theme	software	tech-svcs	transport	advert/retail
as-a-service	2011	**2009**	2013	2012
mobile-app	2010	**2009**	2012	2012
cloud	2012	**2010**	2011	2014
streaming	**2008**	2008	2013	2014
artificial-intelligence	2017	**2016**	2018	2018
augmented/virtual-reality	**2015**	2016	2022	2022
blockchain / crypto	2021	**2018**	–	2022
real-time	**2001**	2003	2009	2014
on-demand	2017	2016	**2014**	2017

Software- and tech-services-origin phrases arrive in the physical-economy classes with lags of one to thirteen years; the only counterexample in the set is on-demand, which originates in transport. The 1% floor is sensitive to class size (transport has roughly one-eighth the volume of software), and the asymmetry is a four-class result that a 45-class build would test properly; the current evidence is consistent with the directional reading but cannot exclude a corpus-construction alternative.

5.2 Theme-level structure

A theme-level analysis with unsupervised topic extraction corroborates the phrase-level pattern without depending on which phrases were picked. This uses its own topic model, fit for theme detection on the four-class granted pool rather than for scoring: LDA with $T = 50$ on a stratified 200,000-filing sample (vocabulary $V = 69{,}361$), transformed across the full 2.32M-filing pool. The resulting topics are recognizable business-model components (health/wellness; autonomous loading and material handling; retail store services; electronic transmission and computer networks; vehicles and traffic).

For each (theme, class) cell that reaches the 1% floor and is tracked at least six years, I fit the Bass diffusion model (Bass, 1969). Of 118 retained fits ($R^2 \geq 0.7$), the median innovation coefficient p is 0.018, the median imitation coefficient q is 0.110, and median $q/p \approx 6.5$, in the range typically reported for consumer-product adoption. Most fits are platform-like; a small tail is fad-shaped (renewable-energy and virtual-reality themes show $q/p > 40$, with steep adoption and visible recession).

Tagging each multi-class theme's origin class (earliest to cross the floor) and counting one directed edge from the origin to each later arrival class gives 72 origin-to-arrival edges over the 33 multi-class themes. Software (009) is the dominant origin, contributing 33 of the 72 edges against the 18 expected under symmetry, and receiving only 10; the software-plus-tech-services pair sends 26 edges to transport and advertising/retail and receives 18 back. The same asymmetry reproduces under independently-fitted NMF on the same sample and vocabulary, with seven of ten curated seed phrases recovering a clear best topic. Both methods are bag-of-words factorizations, so this robustness is within-paradigm; a sentence-embedding or graph-community extraction is the natural cross-paradigm test and is not run here.

6 Do new themes come from incumbents or entrants?

The corpus invites population-share questions, and it punishes them when the base rate is ignored. As a worked example with substantive interest of its own: when a new theme appears, is it carried by new entrants or by incumbents diversifying in? Schumpeter (1942) introduced the distinction (Mark I entrant-led, Mark II incumbent-led), and the standard empirical form reports the entrant share among early adopters.

Applying the standard form to the 50 LDA themes, the 50 earliest carriers per theme are 73.7% entrants on average, and 48 of 50 themes are majority-entrant. The raw numbers look like strong entrant-led adoption. But the corpus's debut rate among all granted filings in this period is 76.4% under year-matched weighting; most filings are by owners debuting that year. Against that baseline the theme-carrier excess is -2.7pp, and 28 of 50 themes sit below baseline. Themes inherit the corpus's high debut rate without adding to it.

The residual pattern is the interesting part. The themes carried disproportionately by incumbents relative to baseline include environmental/sustainability, natural gas and petroleum, artificial intelligence, and cloud computing. The canonically new technology themes of the 2010s and 2020s lean incumbent: established firms diversifying into AI and cloud pull the vocabulary signal forward at least as much as de-novo entrants do. This is consistent with the within-firm patent contrast of Section 4; the firms moving the vocabulary are not necessarily the firms moving the patents.

AI is not (yet) an entrepreneurial explosion

The corpus supports a direct comparison of the AI era to the one episode of genuine vocabulary rupture in the record. Aligning the two eras at their starts (internet 1994, AI 2015) and tracking era-term prevalence in the software and tech-services classes: through year three the two track each other closely (internet 5.1% of filings, AI 4.0%), and then they diverge. Internet vocabulary tripled between years four and six, reaching 23% of filings in 2000; AI vocabulary has compounded steadily to 12.9% at year nine, roughly where the internet stood at year five (Figure 8). The release of consumer chatbots is visible as an acceleration (6.8% in 2022 to 10.2% in 2023), not a discontinuity. Corpus turbulence tells the same story: the per-year mean of prospective surprise, which spiked along with internet vocabulary in the late 1990s relative to its trend, shows no AI-era deviation at all through 2024. Combined with the incumbent lean of the AI theme above, the trademark record's current verdict is that AI is an incumbent-led vocabulary adoption, not an entrepreneurial explosion. The falsifiable forward claim: if AI follows the internet's path with a lag, its prevalence should triple within roughly three years of this paper's data edge, and the entrant share of its carriers should rise above the corpus baseline rather than sit below it.

7 A research agenda the construct opens

Each of the following is computable from the public corpus and panels with the apparatus of this paper. None is run to completion here.

Are transferred ideas as valuable away from home? The flow graph identifies, for every theme, its origin class and arrival classes. Linking carrier filings to the SEC panel tests whether a theme earns the same margin and survival premia in destination industries as in its origin. A first-pass within-firm estimate finds a contemporaneous gross-margin lift of +1.4pp per σ of borrowed novelty ($t = 2.1$) that fades within a year; whether portability is the rule or the exception is open.

Suppliers versus deployers. When a phrase crosses industries, the carrier filings split into firms whose offer *is* the new thing and firms whose offer *supplies* the firms doing the new thing, the sellers of pick-axes rather than the panners for gold. The goods/services text is rich enough to separate the two roles, and the gold-rush intuition that suppliers outperform deployers is testable against renewal, margins, and listing.

Demographic concentration and where themes are born. Surname-imputed demographic composition of filing owners, applied to US-domiciled individual filers, identifies industry enclaves in which a demographic group is heavily over-represented. Preliminary work finds enclave filings sit on the innovation tail of ΔKL rather than the harvest tail, and that owners who diversify across classes list publicly at many times the single-class rate. Whether enclave industries disproportionately originate themes, and whether enclave-rooted firms carry them outward, is a direct join of the demographic panel to the flow graph.

External codification. The USPTO ID Manual periodically standardizes acceptable goods/services wording. If high-ΔKL filings use terms the Manual later codifies, while those terms are still rare in class, the construct anticipates institutional recognition, the strongest available external validation and one that requires only the Manual's version history.

Cross-paradigm theme extraction. The diffusion structure has been shown method-stable within bag-of-words factorizations. A sentence-embedding clustering or graph-community extraction on the same corpus would establish (or refute) paradigm independence.

8 Scope and limits

The paper makes four substantive claims. ΔKL, computed as in Barron et al. (2018); Murdock et al. (2017) on topic distributions of USPTO trademark filings, produces interpretable structure on commercially-incentivised legal text. It marks commercial risk at the mark level, robustly across scoring choices and cohorts: forward-leaning applications complete registration slightly less often (an inverse-U with the flux-neutral middle following through best), and forward-leaning registered marks fail the Section 8 use-proof gates more often — event-dated, +3.9pp first-gate failure across quintiles on 4.12M registrations 2002–2018, positive in 31 of 44 industries, emerging with the 2008–2010 cohorts, buffered by representation and by recent proof of use, attenuating but persisting at the year-ten renewal; the snapshot single-cohort estimate agrees and is scoring-robust (−5.5pp survival under topic scoring at two resolutions and under term scoring). It is empirically distinct from patent-track invention within firm. And it supports direct measurement of cross-industry vocabulary diffusion, with multi-year origin-to-arrival lags, Bass-fittable adoption, and strongly asymmetric flow out of software.

The paper rejects three readings the evidence does not support. ΔKL does not measure innovation in any strong sense; the defended primitive is lexical resonance on commercial text. ΔKL does not identify an exploitable extreme-tail return premium; the large debut-specification figure fails a sample-size diagnostic. And the term-scored firm-level advantages of forward-leaning filings (margins, listing) are not asserted as properties of firms: they dissolve or reverse under distribution scoring at every resolution tried and under document-length controls, and the paper reads them as drafting-style artifacts pending evidence to the contrary.

The gate results are event-dated across all elapsed cohorts, so the earlier single-cohort identification concern is resolved; what remains open is *why* the penalty emerged with the 2008–2010 cohorts (candidate explanations: the post-crisis composition of forward bets, the self-filing boom, the app-era expansion of speculative tech filings) and whether the per-class gradient tracks production lead times or vocabulary churn. The prosecution-telemetry results (response latency, death-stage timing) are estimated on the software diagnostic class; corpus-wide estimation is mechanical now that the event table exists. The theme-level diffusion analysis is a four-class build; per-class ΔKL scores now exist for all 45 classes, but the theme panel and flow graph have not been re-estimated on the full set. The legacy expert-list discriminant ($+0.34\sigma$ on BCG/MIT lists) is withdrawn: it does not replicate on the current pipeline under any scoring ($+0.08$ to $+0.14\sigma$, all $p > 0.05$, $n = 41$ matched firms).

Questions owed to follow-up work: the agenda items of Section 7; whether the mark-level gate penalty reflects vocabulary position itself or selection on owner covariates; the cohort-emergence question above; length-controlled re-estimation of the firm-level panels and the never-registered-applications test of Section 3.3; and corpus-wide estimation of the months-scale freshness measure and the prosecution telemetry.

9 Reproducibility

Code, firm-year panels, and analysis outputs are available at `https://github.com/ericsilver/novelty`. The pipeline rebuilds from a USPTO Open Data Portal API key with a single `make tm` command (the underlying TRTYRAP backfile is approximately 12 GB streamed once). The published panels include the SEC-matched firm-year ΔKL panel (n = 59,033) and the PatentsView-joined panel (n = 174,569). Analysis scripts and their outputs are indexed in the repository `README.md`.

A The outcome analysis without registration conditioning

Figure 9 repeats the lifecycle outcome analysis without the registration conditioning (term-scored, where the composite shape was originally encountered), on filings 2013–2015 pooled across all classes (a window whose registrations land by ∼2017 and face at least one Section 8 gate by the 2025 snapshot; the earliest registrations in the window also face their year-ten renewal, so panel B mixes one and two gates). Panel A is the unconditional composite, the probability that a filing both registers and passes the gate, measured over all filings including the never-registered. It is an inverse-U in ΔKL (quintiles 22.0%, 22.9%, 23.0%, 22.5%, 21.4%; middle-versus-tails depth +1.4pp, SE 0.1pp), because the composite inherits the registration step's middle-favoring shape from Figure 3A. Unconditional survival analyses on this corpus therefore recover an inverse-U whether or not the maintenance gate itself favors the middle, which is why the body of the paper conditions on registration throughout. Panel B shows the conditional gate outcome on the same filings for contrast; on this mixed one-to-two-gate cohort it is mildly inverse-U with a clearly penalized forward tail, consistent in the right tail with the cleaner single-gate cohort of Figure 4 and a reminder that the left tail of the gate result is cohort-sensitive.

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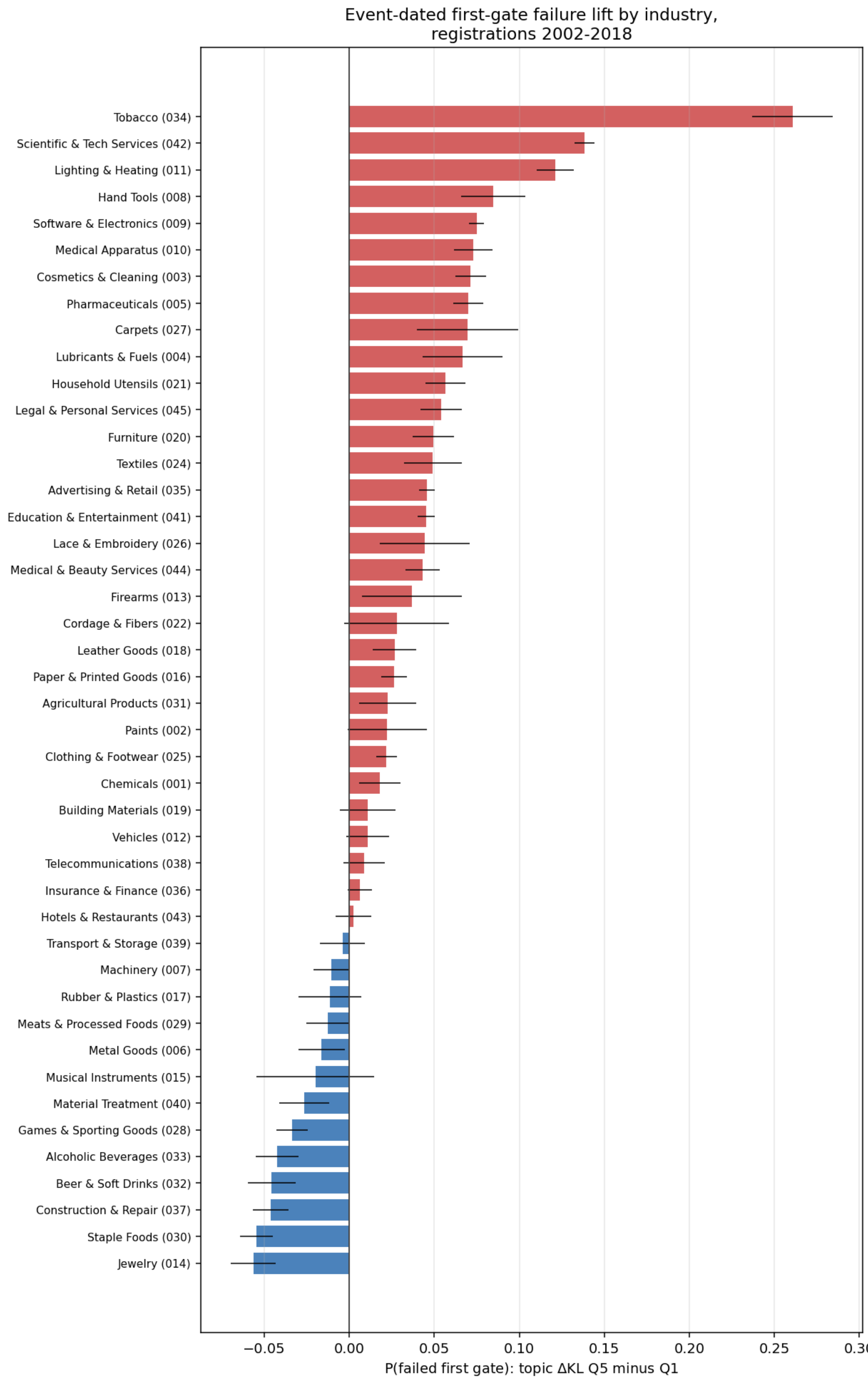

Figure 5: Event-dated first-gate failure lift (topic ΔKL top minus bottom quintile) by industry, registrations 2002–2018, with 95% intervals. Positive bars indicate the forward-leaning penalty.

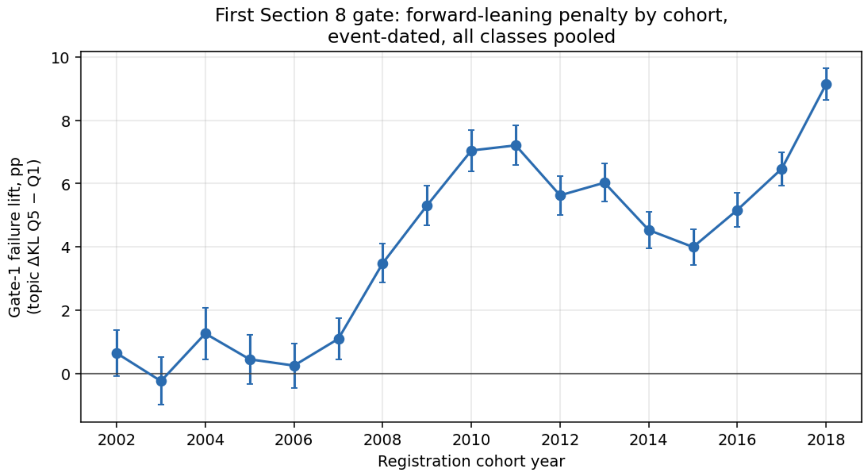

Figure 6: The forward-leaning gate penalty by registration cohort, event-dated, all classes pooled. The penalty emerges with the 2008–2010 cohorts and persists through every cohort since.

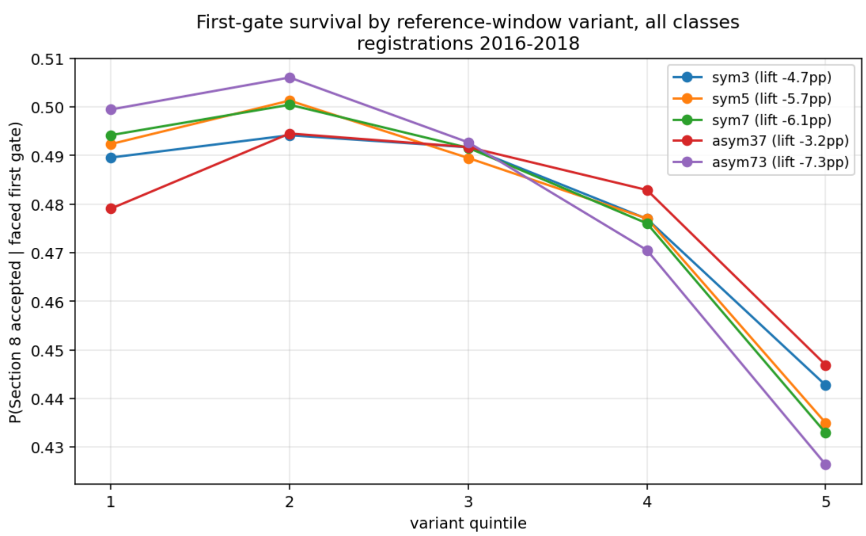

Figure 7: First-gate survival by quintile of each reference-window variant, pooled across classes, registrations 2016–2018. The penalty deepens monotonically from the foresight configuration (pros W=3, retr W=7) to the past-rupture configuration (pros W=7, retr W=3).

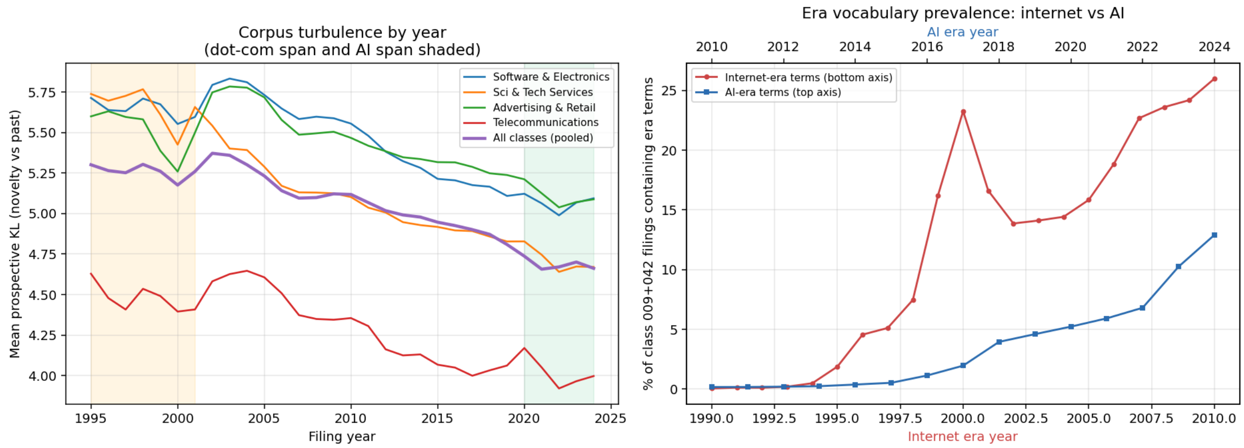

Figure 8: *Left:* per-year mean prospective KL by class and pooled, 1995–2024, with the dot-com and AI spans shaded. *Right:* era-term prevalence in classes 009+042, internet era (bottom axis, from 1994) against AI era (top axis, from 2015).

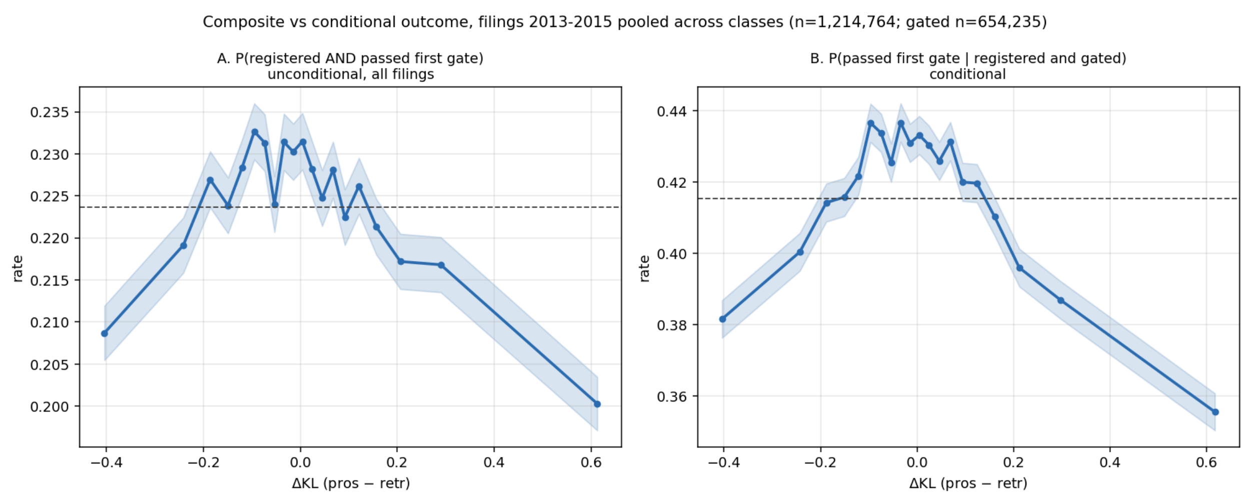

Figure 9: Filings 2013–2015 pooled across classes ($n = 1{,}214{,}764$; gated subset $n = 654{,}235$). *A:* the unconditional composite P(registered and passed first gate) over all filings. *B:* P(passed | registered and gated) on the same filings.